

Tossaporn (Tree) Saengja

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SHORT BIO

Five years' experience in research, industry, and teaching. Developed advanced models and led teams.

EDUCATION

Massachusetts Institute of Technology

Masters of Engineering in Computer Science and Engineering 4.8/5.0 2019 – 2020

Bachelor of Science in Computer Science and Engineering 4.8/5.0 2015 – 2019

AWARDS

4th, MIT ZERO, ACM Boston Preliminary 2016 (BOSPRe) among 37 teams over 16 schools

85th, Bronze, **International Olympiad in Informatics 2013 (IOI)** among 299 contestants over 77 countries

EXPERIENCE

PreceptorAI, CARIVA — AI Consultant

2023 – Present

- Developed and deployed, in collaboration with Siriraj Radiology Department, a multi-label (muscle, fats) segmentation model (nnUNet, 0.98 Dice) for vertebrae L3 in MRI images (Pytorch).
- Developed, in collaboration with Siriraj Radiology Department, an image-to-text model (Vision-Large Language Model) on chest X-ray images to report findings (Pytorch).
- Developed, in collaboration with Siriraj Radiology Department, a generative text-to-image model (diffusion model) for chest X-ray images (Pytorch).

Vision and Learning Lab, VISTEC — Research Assistant

2023 – 2024

- Under supervision of Prof. Supasorn Suwajanakorn, working on generative models, especially diffusion models.
- Worked on collocation problem of real images by distilling prior knowledge (Score Distillation Sampling) from the stable diffusion model.
- Experimented on memorization of diffusion models with CIFAR-10 dataset.

Mahidol Wittayanusorn High School — Computer Science Teacher

2020 – 2023

- Coached students to represent Thailand in the IOI 2022.
- Taught multiple subjects including data science (Python), special topics (HTML, CSS, React, Blockchain), Computer Olympiad (C++), and basic Python.
- Advised twelve year-long projects, mostly in the machine learning area.

Agnos — Technical Cofounder

2019 – 2023

- Worked with physicians to develop a medical question system for diagnosis (Python, Django) (130k diagnosis records, 50k active users).
- Led multiple teams: Data Science (Redash), AI, Growth.
- Team raised funds in Shark Tank Thailand S3 for 30M baht.

Facebook, Notification Backend — Intern

2017

- Implemented a production pipeline for network effect prediction models, both classification and regression, using HiveQL and FBLeanner Flow within Facebook's machine learning framework.
- Increased 0.5% comments metrics in production with 2 billion active users at the time.

MIT Media Lab, Human Dynamics — Researcher

2018 – 2020

- Under supervision of Yan Leng, developed a novel optimization framework to learn the underlying network structure and individual marginal benefits in large-scale, real-world datasets.
- Learning Quadratic Games on Large-Scale Networks: Prepared large-scale, real-world datasets for optimization frameworks to learn the underlying network structure and individual marginal benefits.

PUBLICATIONS

Udomlaksakul, K., Pengpun, P., Saengja, T., Veerakanjana, K., Tiankanon, K., Khlaisamniang, P., Supholkhan, P., Chinkamol, A., Aussavavirojekul, P., Phimsiri, H., Sripo, T., Boonnag, C., Tongdee, T., Siriapisith, T., Saiviroonporn, P., Kinchagawat, J., & Ittichaiwong, P. (2024, August). GPU Poor's Guide to Radiology Report Generation. *The 23rd Workshop on Biomedical Natural Language Processing and Bionlp Shared Tasks*.